

Solve each equation. Show your work in the space provided.

1. $-5(7 - 2y) + 3(4y + 1) = 12$

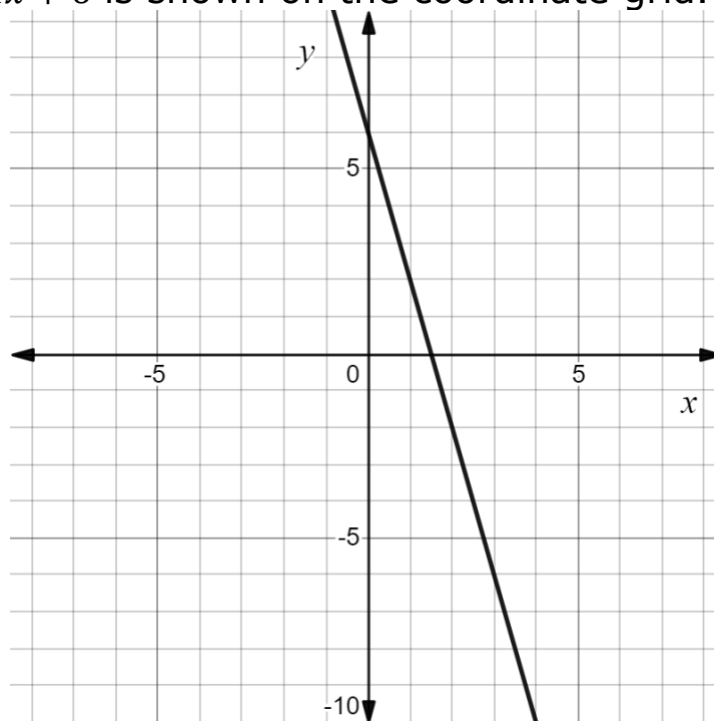
2. $6.1(14 - 3h) = -7.9h + 59.4$

3. $\frac{7}{5}(-2 + 8k) = \frac{3}{5}(2k + 22)$

4. $6(4r + 3) = 8(2 + 3r) + 2$

5. $\frac{1}{3}(2x - 54) + 4\frac{1}{3}x = -7$

6. The line $y = -4x + 6$ is shown on the coordinate grid.



Draw a line on the coordinate grid so that the solution to the system of equations graphed on the grid would be “no solution”.

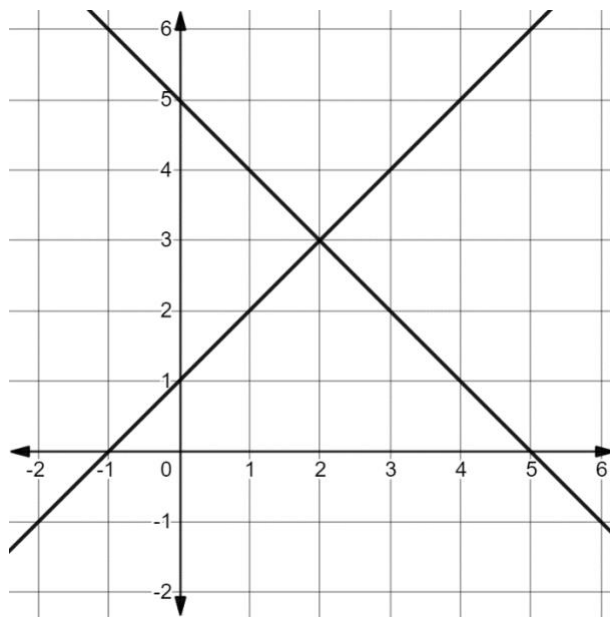
7. A system of equations is shown.

$$\begin{cases} y = -\frac{9}{4}x + 2 \\ y = 3x - 19 \end{cases}$$

Determine which, if any, of the ordered pairs are solutions to the system. Show your work in the provided space.

Ordered Pair	Work	Solution?
a. (8,5)		☐ Yes ☐ No
b. (4, -7)		☐ Yes ☐ No
c. (-4,11)		☐ Yes ☐ No

8. The graph of the system of equations $y = -x + 5$
 $y = x + 1$ is shown.



Complete the statement.

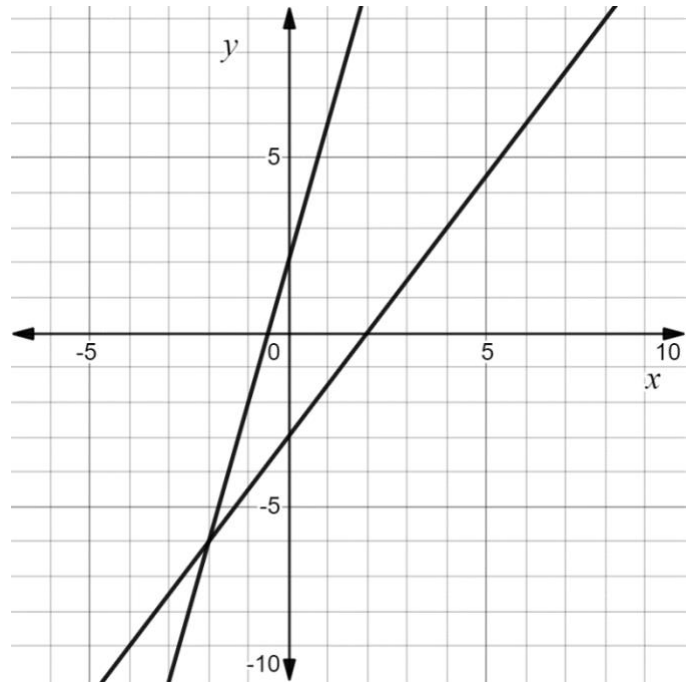
The point (2,3) is

- ☐ the slope
- ☐ a solution
- ☐ the y-intercept

of

- ☐ $y = x + 1$ only.
- ☐ $y = -x + 5$ only.
- ☐ $y = -x + 5$ and $y = x + 1$.

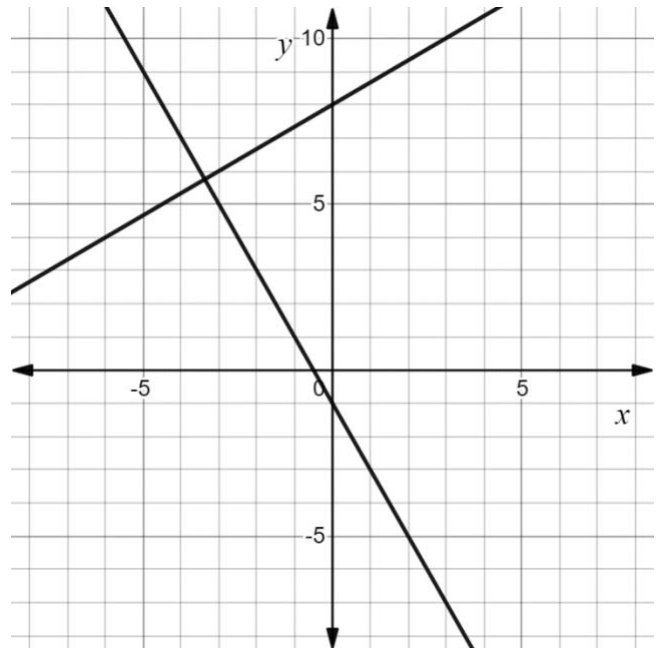
9. The graph of the system $\begin{cases} y = \frac{3}{2}x - 3 \\ y = 4x + 2 \end{cases}$ is shown.



What is the solution to this system?

Solution	
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10. The graph of the system $\begin{cases} y = -2x - 1 \\ y = \frac{2}{3}x + 8 \end{cases}$ is shown.

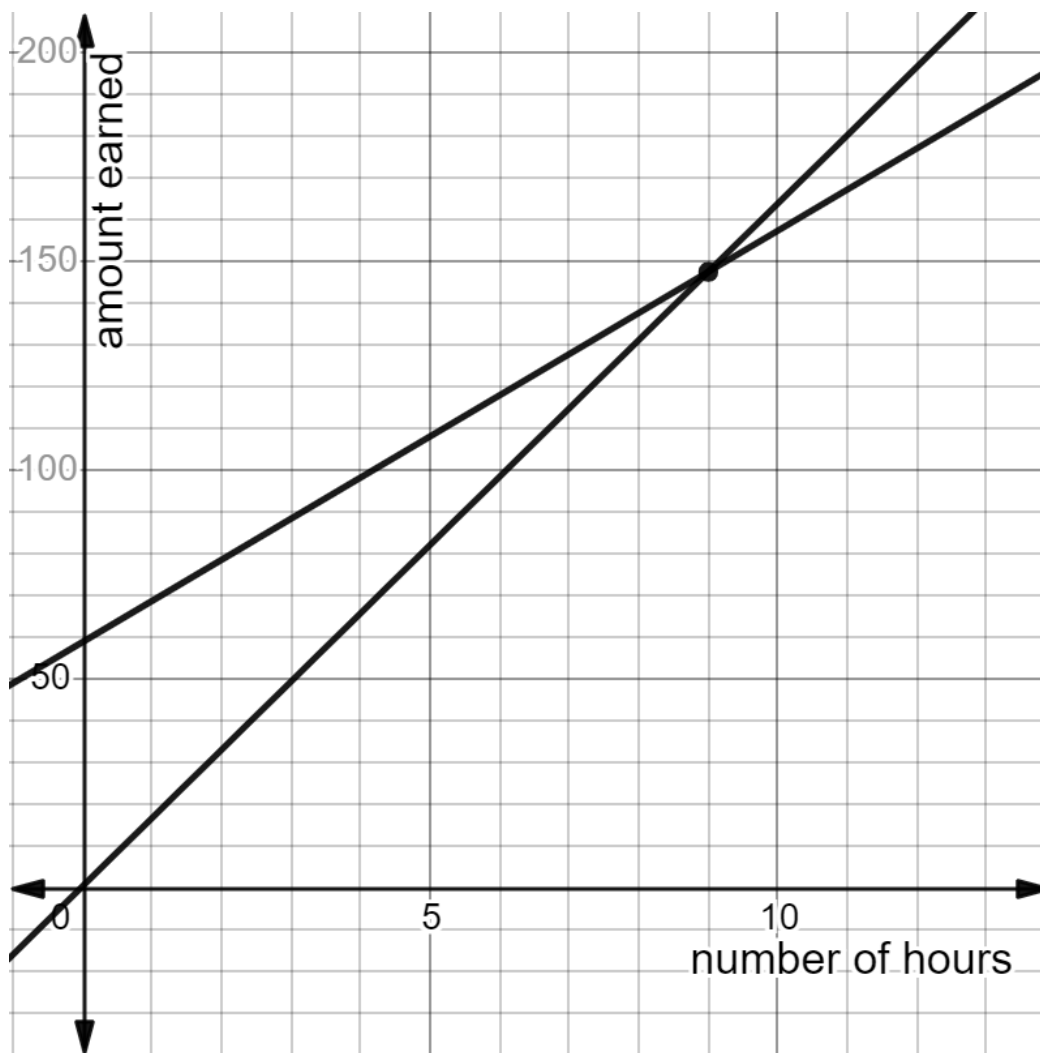


What is the approximate solution to this system?

Approximate Solution	
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11. Serena just started working at the local salon. She is paid \$9.85 per hour and earns, on average, \$58.50 in tips each day. Roseline just started working at a bank. Since she does not receive tips, she earns \$16.35 an hour. The system of equations $\begin{cases} y = 9.85x + 58.50 \\ y = 16.35x \end{cases}$ can be used to represent the amount of money earned, y , each day for working x hours.

The graph of the system is shown.

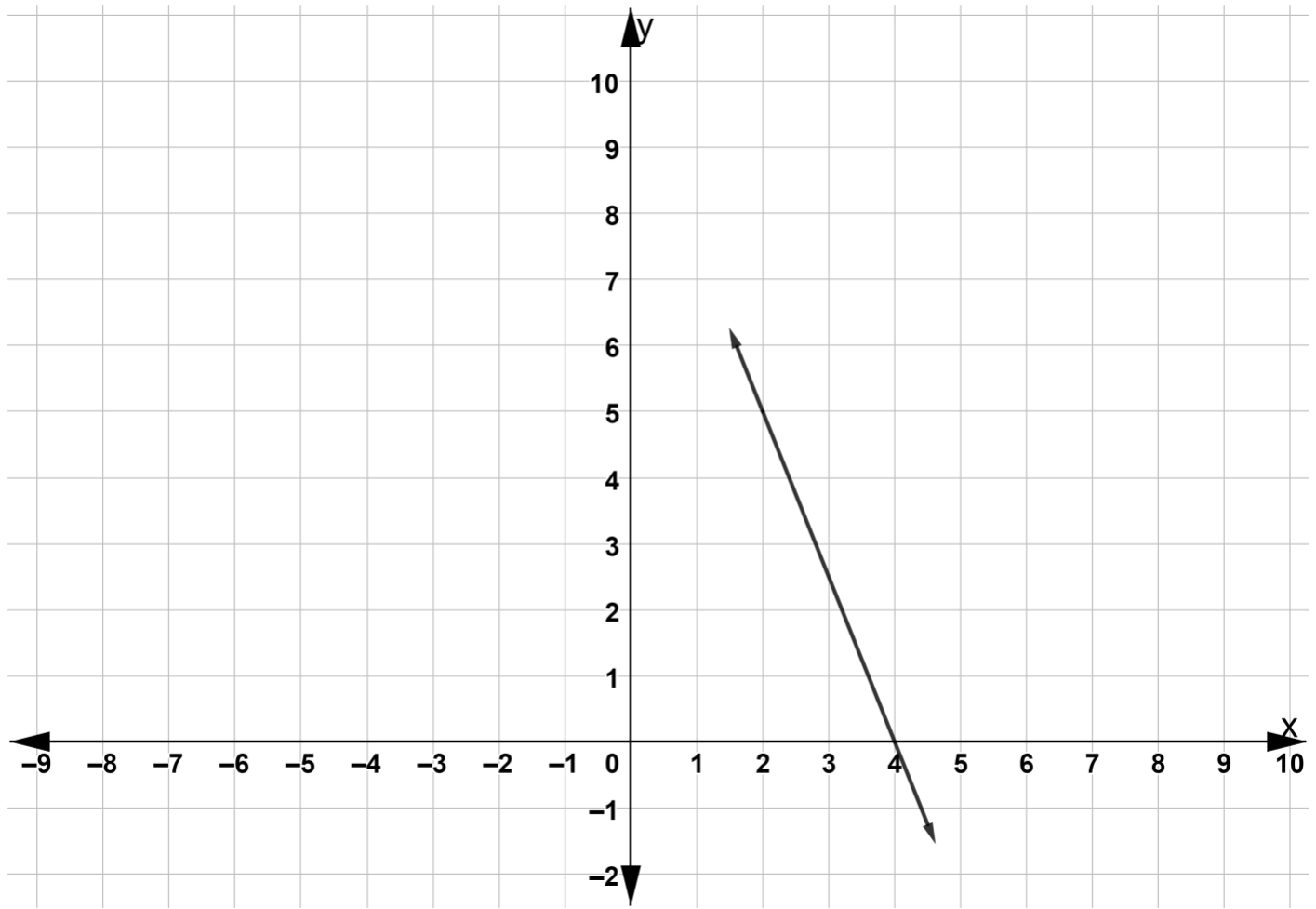


What is the approximate solution to this system?

Approximate Solution	
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12. The equation $y = 10 - 2.5x$ represents the amount of water, y , in gallons in Tank 1 after x seconds. The equation $y = -\frac{5}{2}x + 10$ represents the amount of water, y , in gallons in Tank 2 after x seconds.

A coordinate grid is shown, where the equation $y = 10 - 2.5x$ is graphed.



- a. Graph $y = -\frac{5}{2}x + 10$ to complete the graph of the system $\begin{cases} y = 10 - 2.5x \\ y = -\frac{5}{2}x + 10 \end{cases}$.
- b. Complete the statements.

The two equations in the system are lines with

- ☐ the same slope.
- ☐ different slopes.

The system $\begin{cases} y = 10 - 2.5x \\ y = -\frac{5}{2}x + 10 \end{cases}$

- ☐ has no solution.
- ☐ has one solution.
- ☐ has an infinite number of solutions.